Ruodi (Tinah) Yuan, BASc

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Seeking interships from May 2027 to Sept 2027

Education

University of Toronto St. George — Toronto, Canada
BASc in Mechanical Engineering ***+ PEY Co-op*** — *Sept 2025 – Apr 2030 (expected)*

University of Toronto Scarborough — Toronto, Canada
Honours Bachelor of Science; Life Sciences (Health Sciences Stream); cGPA: 3.90/4.00 | University of Toronto Scarborough Dean's List — *Sept 2024 – Apr 2025*

Functional Experience

- Designed and validated Formula SAE drivetrain systems using SolidWorks for 3D modeling and assembly integration; supported AWD EV architecture development, PM motor assembly, rotor alignment tooling, and mechanical packaging of motor mounts, shafts, bearings, and cooling hardware.
- Developed simulation and analysis workflows using MATLAB, Simulink, and ANSYS; digitized motor efficiency maps, built lookup-table tools, evaluated endurance-event heat loads, and validated drivetrain and cooling components under representative operating conditions.
- Produced GD&T-compliant drawings, cost analysis, and manufacturing documentation for machined drivetrain components; applied DFM/DFA, tolerance planning, supplier-ready drawing practices, and fabrication constraints to support machining and assembly.
- Designed and iterated aerospace and robotic mechanisms using SolidWorks, including payload pickup/release systems, landing gear integration concepts, custom robot housings, and 3D-printed prototypes validated through hands-on testing and refinement.

SAE Experiences

University of Toronto Formula SAE Racing — Toronto, Canada
Vehicle Dynamics Lead — *June 2026 - Present*
- Elected as Vehicle Dynamics Lead, responsible for developing simulation tools to evaluate system trade-offs across suspension, steering, brakes, chassis, and powertrain.
- Responsible for supporting suspension pickup point, kinematics, and compliance target decisions through integrated vehicle modelling.
- Expected to coordinate with Race Engineering on testing plans, vehicle tuning, driver-in-the-loop validation, and tire performance optimization.

Drivetrain Team Member → Drivetrain Team Lead — *Sept 2024 - June 2026*
- Co-led drivetrain architecture development for the team's first AWD Formula SAE EV race car and performed PM motor assembly.
- Identified rotor alignment as a key challenge in PM motor assembly due to strong magnetic attraction forces between the rotor and stator; researched industry assembly practices and designed a custom alignment tool to enable safe, precise, and repeatable rotor installation.
- Produced GD&T-compliant mechanical drawings for the inboard motor mount, planet shafts, and motor bearing, along with cost analysis and manufacturing process documentation, to support accurate fabrication and machining.
- Designed a comprehensive dual inverter cooling system using Simulink, optimizing thermal management for enhanced performance and efficiency, with SolidWorks for CAD and validated components with ANSYS FEA to ensure structural integrity under operating loads.
- Sized the system from a heat load estimate by digitizing the motor's power-efficiency map (Fischer) into a high-resolution lookup table using automated MATLAB tools and integrating it into a Simulink vehicle model to evaluate motor efficiency continuously through a simulated endurance event, establishing the thermal boundary condition for the cooling system simulation.
- Participated in Formula SAE Michigan 2026, delivering design presentations on drivetrain architecture and engineering decisions to industry judges as part of the competition's design judging event.

University of Toronto Aerospace Team — Toronto, Canada
Unmanned Aerial Systems – SAE Payload Specialist — *June 2026 - Present*
- Responsible to lead the design of an autonomous moving payload for for the SAE Aero Design Advanced Class mission, focusing on lightweight mechanical systems and reliable autonomous operation.
- Developing navigation, capture, and release mechanisms through CAD, rapid prototyping, and engineering trade studies to maximize mission reliability.

Unmanned Aerial Systems – SAE Structure Member — *Sept 2025 - June 2026*
- Designed and iteratively refined payload pickup and release mechanisms for a hybrid V/STOL aircraft using SolidWorks, validating designs through 3D-printed prototypes to improve landing accuracy and reduce mechanical instability during deployment
- Collaborated cross-functionally with control and avionics teams to align mechanical design with design requirements.
- Contributed to landing gear design and fuselage structural integration, designing for accessibility to simplify maintenance and gaining hands-on experience in composite and mechanical assembly.

Additional Experiences

UofTHacks 13 (Project Link) — Toronto, Canada
2nd Place – Hack the Human-Robot Experience — *Jan 2026*
- Built Red Lamp, a companion robot for students addressing isolation and burnout by providing emotionally aware encouragement during independent study sessions.
- Integrated hardware systems using a LeLamp kit, Raspberry Pi, sensors, LEDs, and custom 3D-printed components, transforming a standard robot into a responsive, companion-focused device.
- Debugged and optimized hardware–software interaction by troubleshooting power instability, loose wiring, and faulty components through voltage testing and iterative assembly, achieving reliable real-time responsiveness.
- Adapted system architecture by incorporating Arduino components after Raspberry Pi connectivity failures, improving system stability and overall performance under real-world constraints.

2025 Engineering Business Future Case Competition — Toronto, Canada
3rd Place Winning Team — *Oct 2025*
- Worked 3 other engineering students to develop a marketing campaign for Vick's Early Defense Nasal Spray, aiming to increase household penetration by 10% in a year.
- Identified a key gap – lack of exposure and reliability – and built strategy around tackling it using frameworks such as Porter's Five Forces and PESTLE.
- Delivered an evidence-based presentation that impressed judges with its clarity, feasibility, and analytical depth.

Make UofT Hackathon 2026 (Project Link) — Toronto, Canada
Participant — *Feb 2026*
- Built an autonomous AI robot using Raspberry Pi, OpenCV, and Arduino for real-time face detection, navigation, and expressive interaction.
- Integrated hardware and software systems, including motor control, servo-driven head tilting, dual power management, and a smartphone-based animated display.
- Optimized vision and motion performance, resolving networking issues and improving detection accuracy, and developed both a working prototype and a 3D-printable final design.

Technical Skills

- **Programming & Tools:** Python, C, C++, Arduino, JavaScript, HTML, MySQL, Microsoft Office Suite.
- **Engineering & Technical:** SolidWorks, MATLAB, Simulink, Catia, ANSYS, machining and manufacturing processes, Vi-Grade CarRealTime, rFpro, ADAMS Car.
- **Language:** English(native), Chinese(native), Japanese(elementary).

Licenses & Certifications

Certified SOLIDWORKS Associate (CSWA) – ***Issued by*** **Dassault Systèmes** — Dec 2025
Basic Machining: Lathe, Mill, Drill Press – ***Issued by*** **George Brown College** — Oct 2025
CPR & First Aid – ***Issued by*** **Canadian Red Cross** — Sept 2024